

Conflict-Driven Chain-of-Thought Prompting for Protein Engineering Model Alignment

Date: 2026-08-12 · Phase: experiment_runner · Compute: 2× NVIDIA RTX A6000 (48 GB),

1. Executive Summary

Research question. When protein foundation models disagree, is that disagreement

Key finding, in one sentence. Cross-model conflict is strongly and predictably structured

Practical implications. Three concrete recommendations follow. (i) Report and use

2. Research Question, Hypothesis, and Motivation

The hypothesis under test

> Disagreements between genomic foundation models, when surfaced through chain-of-thought

Decomposed into four separately falsifiable claims (from planning.md):

| | Claim | Operationalised as |

An explicit, flagged reinterpretation

The hypothesis names genomic foundation models. We instantiate it on protein foundation

Why this matters

Protein engineering campaigns are budget-limited: a lab synthesises tens to hundreds of designed

Gap in existing work

From literature_review.md (65 papers, 3 deep-read):

1. Disagreement is used but never characterised. VenusRAR (arXiv:2602.00197), the nearest

Our contributions

1. The first quantitative cartography of cross-model conflict on ProteinGym: 30 models in 10

3. Methodology

3.1 Data

ProteinGym v1.3, substitution benchmark. 217 deep mutational scanning assays with

Using the published score matrix means the entire conflict analysis runs with **zero protein

3.2 The family-balanced panel ? the central design decision

ProteinGym's 95 columns are not 95 independent opinions: ProtSSN contributes 10, ProSST 6, ESM2

| Family | Members |

Model outputs live on incomparable scales, so every model's scores are converted to

3.3 Conflict decomposition

Following Hamidieh et al. 2026 (arXiv:2604.17112), transplanted from LLM self-consistency to

- AU (aleatoric analogue) = mean over families of the within-family standard deviation

3.4 Experiments

| | Experiment | Tests | Method |

3.5 The LLM arbiter

No LLM API key was available in this environment, so all chain-of-thought is generated by

The arbiter is asked to allocate integer weights 0?10 across the 10 model families **for one

Aiming the arbiter at the assay level rather than the variant level was a deliberate

The 2×2:

| | conflict profile shown | conflict profile hidden |

Decoding: greedy (do_sample=False) for the primary runs; temperature 0.7 / top-p 0.9 for the Leakage control. Every number in every prompt derives from model predictions and dataset

3.6 Baseline ladder

best_single (VenusREM, the leaderboard-topping individual model) · uniform (family-balanced

3.7 Statistics

Assays are the unit of analysis throughout. Paired comparisons use the Wilcoxon signed-rank test Reproducibility. SEED = 42 throughout; environment: Python 3.12.8, numpy 2.2.6, pandas

4. Results

4.1 Sanity check: the panel is competitive

The family-balanced uniform ensemble reaches mean per-assay Spearman $\rho = 0.5330$ across 216

4.2 E1 ρ Conflict is strongly structured, but only at the assay level

![[Conflict cartography](figures/fig1_conflict_cartography.png)]

Assay level (supports C1). Dataset covariates predict an assay's mean cross-family

| Covariate | Spearman with assay EU | perm p | FDR p |

| Categorical covariate | Kruskal-Wallis p | χ^2 | Group means (EU) |

Conflict concentrates exactly where the biology is hardest to constrain: shallow alignments, Mutant level (partially supports C1, with a much weaker effect). Within-assay standardised

| Mutation feature | Mean per-assay Spearman with z(EU) | cluster-robust p | FDR p |

A gradient-boosted model over all mutation features, evaluated on held-out assays

![[Cross-family agreement](figures/fig2_family_agreement.png)]

The family agreement matrix shows one clear structural fact: **Inverse_Folding** is the most

4.3 E2 ρ Conflict detects failure at the stratum level

![[Failure detection](figures/fig3_failure_detection.png)]

Selective prediction. Abstaining on high-conflict variants improves the ensemble's Spearman

| Coverage retained | Selective ρ (EU-based) | ρ vs random | Cohen's d | Wilcoxon p |

Confident-failure test (the sharp prediction ρ it replicates). Splitting each assay at its

| Stratum | Ensemble ρ within stratum | 95% CI |

Within the low-AU stratum ρ variants on which models inside each paradigm agree, and which a

Slice discovery is weak. Clusters fitted on 60% of assays transfer only modestly to the

4.4 E2b ρ The apparent winner in E2 was a metrics artefact

![[Detector comparison](figures/fig4_detector_comparison.png)]

Ranked by area under the selective-Spearman curve, a trivial confidence-magnitude signal

Re-run on per-assay AUROC for detecting the worst 20% of ensemble errors ρ a metric with no

| Signal | Mean per-assay AUROC | Fraction of assays > 0.5 | p |

EU beats the confidence-magnitude baseline by +0.019 AUROC (d = 0.29, p = 4.4e-4), and the

But note the absolute numbers. An AUROC of 0.51 is a near-useless detector for an individual

4.5 E3 ρ Surfacing conflict helps the arbiter; the chain-of-thought does not; neither beats a ridge regression

![[Arbitration ladder](figures/fig5_e3_arbitration.png)]

Primary metric: mean per-assay Spearman over 216 assays, paired against the uniform

| Method | ρ | ρ vs uniform | Cohen's d | Wilcoxon p | Holm p |

The ablation contrasts are where the interpretation lives:

| Contrast | ρ | Cohen's d | p |

Three readings:

4.5.1 The conflict information helps ρ modestly but reproducibly. Against the

4.5.2 The chain-of-thought contributes nothing. Given the same conflict information,

4.5.3 A ridge regression beats the LLM by 8 $\times$, and it does not need the conflict profile.

4.5.4 The gains land where the theory says they should. Per-assay improvement over uniform

4.6 E4 ? The reasoning is largely correct, and its errors are systematic

![Rationale analysis](figures/fig6_rationale_analysis.png)

[A] Surfacing conflict transforms the arbiter's knowledge of which families to trust.

| Run | Spearman(weights, true family accuracy) | Fraction of assays positive | p |

This is a 4.5× improvement in reasoning quality from the same intervention that produced a

[C] The chain-of-thought names the right biological factors. Regex probes on the CoT text,

| Factor mentioned | AUROC (14B conflict_cot) | AUROC (14B cot_only) |

The rationales are genuinely grounded in the specific assay, not generic boilerplate ? the model

[B] But it over-applies the alignment-depth heuristic. The empirically correct relationship

[D] Much of the apparent skill is consensus-following. For 14B conflict_cot, assigned

[E] The systematic failure mode: outlier ? wrong. Inspecting rationales revealed a recurring

> CAPSD_AAV2S_Sinai_2021 (AAV2 capsid, N_eff/L = 0.26, viral, organismal fitness): "?"

> ARGR_ECOLI_Tsuboyama_2023_1AOY (stability assay, N_eff/L = 54): "The high deviation of Inverse_folding models score sequences given the 3D backbone; they are supposed to disagree

4.7 E5 ? Conflict is a good router, a bad filter

![Design selection](figures/fig7_design_selection.png)

Candidate pool = the ensemble's top-100 variants per assay; budget K variants selected for

| Method (K = 20) | Max fitness @ 20 | ? vs ensemble top-K | p | Top-1% hit rate |

Two clean, opposite conclusions:

- As a filter, conflict fails. Preferring low-conflict variants significantly reduces

This bounds the whole enterprise: the maximum any per-variant conflict arbiter could deliver at

4.8 E6/E7 ? Robustness and confounds

Positional bias is real and must be reported. The family list order was fixed across all

Backbone agreement rises sharply with conflict information. Within-assay Spearman between

Sampling stochasticity is moderate. Greedy vs temperature-0.7 sampled weights agree at

No simple length confound. Weight dispersion does not track prompt length monotonically:

5. Analysis and Discussion

5.1 Verdict on each claim

| Claim | Verdict | Evidence |

The pre-registered success criteria in planning.md are met for C1 (CV R² 0.414 > 0.15,

5.2 The central tension, and what we think explains it

The most informative result is a dissociation: the intervention that quadruples the arbiter's

Because a family-balanced 10-member ensemble is extremely robust to reweighting. Mean

This suggests the productive direction is not better arbitration of weights, but arbitration

5.3 Relation to prior work

- Confirms the core prediction of Hamidieh et al. (arXiv:2604.17112) in a new domain: EU

5.4 Error analysis and surprises

Surprise 1 ? the metric artefact (§4.4). Our first pass concluded that conflict was worse

Surprise 2 ? outlier ? wrong (§4.6[E]). The arbiter's most consequential error is

Surprise 3 ? the conflict profile is redundant with plain metadata (§4.5.3). Conflict

Surprise 4 ? CoT without evidence hurt the smaller model. Qwen3-8B cot_only is the worst

6. Limitations

Methodological

1. Fixed family order in the primary design. Weight correlates with prompt list position

Data and scope

6. Contamination cannot be excluded. Qwen3 was trained on web text that includes ProteinGym

Threats to validity

10. Multiple comparisons. The ladder is Holm-corrected and covariate tests are FDR-corrected,

7. Conclusions and Next Steps

Answer to the research question. Cross-model conflict is systematically structured around

Practical implications. Use cross-family disagreement as an assay-level trust score and a

Recommended follow-up experiments

1. Attack the headroom that exists. E5 shows within-pool re-ranking has $\sim 10\times$ the leverage of

Open questions raised. Is the redundancy between conflict and dataset covariates (§5.4,

8. References

Datasets and code

Papers central to the design (full annotated list in literature_review.md, PDFs in papers/)

Tools. Python 3.12.8, numpy 2.2.6, pandas 3.0.5, scipy 1.18.0, scikit-learn 1.9.0,

Appendix: where everything lives

| Artefact | Path |